Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "induction" and "flux" to explain your answer. Sentence starters are optional.

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

emf: energy supplied per charge, measured in volts

01 Warm-up: bring back an earlier idea

A **3 W** device runs for **5 s**. **Find transferred energy**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and **mark the changing field through** the loop. Contrast it with a stationary case.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a magnet held still** beside a fixed coil with a magnet moving into the coil.
2. Which can produce a changing flux?

Sentence starter: Both ... . They differ in ... because ... .

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

04 Apply the physics

A lab group pushes a magnet into a fixed coil and records a positive meter deflection.

1. **Draw the approach** and withdrawal motions.
2. **Predict the sign** when the magnet is pulled back out along the same path without changing connections.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep charge separation fixed and multiply one charge magnitude by **4**. **Compare electric-force magnitude**. **Label charges** and both interaction forces.

Word help: separation = distance between the specified objects or centers

Sentence starter: Both … . They differ in … because … .



Charge A Charge B

Marzano check - circle: 1 2 3

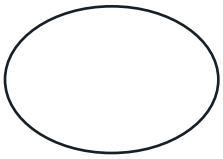
07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

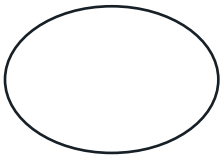
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Predict how faster approach** and more turns can change induced voltage.
2. **State which comparisons must keep** everything else in the setup fixed.

Sentence starter: I predict … . I used … to make this prediction because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3